

Problem 3A.

Prove that $\rho \left(C_p \frac{\partial T}{\partial P} \right)_H = \frac{\partial \ln V}{\partial \ln T}_P - 1$

From $dH = C_p dT + (1 - \beta T) V dP$ where $\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$ we

obtain $C_p \left(\frac{\partial T}{\partial P} \right)_H = (\beta T - 1) V \Rightarrow \frac{1}{V} \left(C_p \frac{\partial T}{\partial P} \right)_H = \beta T - 1 \Rightarrow$

$$\rho \left(C_p \frac{\partial T}{\partial P} \right)_H = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P - 1 = \frac{\partial \ln V}{\partial \ln T}_P - 1$$

For ideal gas $\beta T = 1 \Rightarrow \left(\frac{\partial T}{\partial P} \right)_H = 0$

Problem 3B.

Prove that $C_v \left(\frac{\partial P}{\partial \rho} \right)_S = C_p \left(\frac{\partial P}{\partial \rho} \right)_T \Leftrightarrow C_v \left(\frac{\partial P}{\partial V} \right)_S = C_p \left(\frac{\partial P}{\partial V} \right)_T$

Since $d\rho = -\frac{1}{V^2} dV$

From $U = U(P, V) \Rightarrow dU = \left(\frac{\partial U}{\partial P} \right)_V dP + \left(\frac{\partial U}{\partial V} \right)_P dV$

Also $dU = T dS - P dV \Rightarrow T dS = dU + P dV \Rightarrow$

$$T dS = \left(\frac{\partial U}{\partial P} \right)_V dP + \left(P + \left(\frac{\partial U}{\partial V} \right)_P \right) dV \Rightarrow$$

$$\left(\frac{\partial P}{\partial V} \right)_S = - \frac{P + \left(\frac{\partial U}{\partial V} \right)_P}{\left(\frac{\partial U}{\partial P} \right)_V}$$

From $H = U + PV \Rightarrow \left(\frac{\partial H}{\partial V} \right)_P = \left(\frac{\partial U}{\partial V} \right)_P + P$

and therefore

$$\left(\frac{\partial P}{\partial V} \right)_S = - \frac{\left(\frac{\partial H}{\partial V} \right)_P}{\left(\frac{\partial U}{\partial P} \right)_V}$$

(Eq. 3B-1)

Consider now the functions

$$H = H(T, P) \Rightarrow dH = C_p dT + \left(\frac{\partial H}{\partial P}\right)_T dP \Rightarrow$$

$$\left(\frac{\partial H}{\partial V}\right)_P = C_p \left(\frac{\partial T}{\partial V}\right)_P \quad (\text{Eq. 3B-2})$$

$$U = U(T, V) \Rightarrow dU = C_v dT + \left(\frac{\partial U}{\partial V}\right)_T dV \Rightarrow$$

$$\left(\frac{\partial U}{\partial P}\right)_V = C_v \left(\frac{\partial T}{\partial P}\right)_V \quad (\text{Eq. 3B-3})$$

Now combining Eq. (3B-1), (3B-2), (3B-3) we obtain

$$\left(\frac{\partial P}{\partial V}\right)_S = -\frac{C_p}{C_v} \frac{\left(\frac{\partial T}{\partial V}\right)_P}{\left(\frac{\partial T}{\partial P}\right)_V} \quad (\text{Eq. 3B-4})$$

Now consider the function $T = T(P, V) \Rightarrow$

$$dT = \left(\frac{\partial T}{\partial P}\right)_V dP + \left(\frac{\partial T}{\partial V}\right)_P dV \Rightarrow \left(\frac{\partial T}{\partial P}\right)_V \left(\frac{\partial P}{\partial V}\right)_T + \left(\frac{\partial T}{\partial V}\right)_P = 0 \Rightarrow$$

$$\frac{\left(\frac{\partial T}{\partial V}\right)_P}{\left(\frac{\partial T}{\partial P}\right)_V} = -\left(\frac{\partial P}{\partial V}\right)_T \quad (\text{Eq. 3B-5})$$

Combining Equations (3B-1) & (3B-5) we obtain

$$C_v \left(\frac{\partial P}{\partial V}\right)_S = C_p \left(\frac{\partial P}{\partial V}\right)_T \quad \left\{ \begin{array}{l} u^2 = \left(\frac{\partial P}{\partial V}\right)_S = -v^2 \left(\frac{\partial P}{\partial V}\right)_S \Rightarrow \\ \left(\frac{\partial P}{\partial V}\right)_T = -\frac{RT}{V^2} \end{array} \right.$$

For ideal gas $\left(\frac{\partial P}{\partial V}\right)_T = -\frac{RT}{V^2}$

$$u^2 = -V^2 \left(-\frac{RT}{V^2}\right) \frac{C_p}{C_v} \Rightarrow u^2 = \gamma RT \Rightarrow u = \sqrt{\gamma RT} \quad \gamma = \frac{C_p}{C_v}$$

Problem 3C.

As a first step we must calculate the required cooling for the natural gas stream. Using our CHE301 calculator we obtain

$$H_5 (@ 25^\circ\text{C}) = -74901.97 \text{ kJ/kmol}$$

$$H_6 (@ 37^\circ\text{C}) = -76998.56 \text{ kJ/kmol}$$

$$\text{The natural gas flow rate is } 100 \text{ MMSCFD} \times \frac{0.0138346 \text{ kmol/s}}{\text{MMSCFD}} \Rightarrow$$

flow rate = 1.3835 kmol/s . Therefore the required duty for the cooling is $1.3835 \cdot (-76998.56 + 74901.97) \Rightarrow$ required cooling -2900.55 kW

From the process flow diagram we see that S3 is saturated liquid @ 35°C and the stream S1 is saturated vapor @ 1 bar. Using our calculator we obtain

$$H_1 (@ 1 \text{ bar}) = -108364.9 \text{ kJ/kmol}$$

$$H_3 (@ 35^\circ\text{C}) = -117027.6 \text{ kJ/kmol}$$

Since the valve is adiabatic the duty of the chiller must be the same as the cooling duty for the natural gas, therefore the flow rate of the refrigerant is

$$F = \frac{-2900.55}{(-117027.6 + 108364.9)} \Rightarrow F = \underline{\underline{0.3348 \text{ kmol/s}}}$$

The mass flow rate is kmol/s is

$$F = 0.3348 \times 44.097 \times \frac{3600}{1000000} \times 24 \times 0.9 \times 365 \Rightarrow \underline{\underline{F = 419 \text{ KMTA}}}$$

The duty of the chiller has already been calculated

$$\dot{Q}_{\text{chiller}} = \underline{2900.55 \text{ kW}}$$

To calculate the duty of the condenser we must first figure out the compressor power requirements.

Case 1. Isentropic efficiency $\eta = \frac{W^{\text{ideal}}}{W} = 0.75$

The discharge pressure of the compressor is the saturated vapor pressure @ 35 °C. Using our calculator we obtain $P_2 = \underline{11.44 \text{ bar}}$

First let's calculate the ideal work.

Step 1 calculate temperature of the compressor discharge stream by solving the entropy balance $\Delta S = 0$ for T_2^{ideal}

The enthalpy & entropy of stream (S1) are

$$H_1 (@ 1 \text{ bar}) = -108,364.9 \text{ kJ/kmol}$$

$$S_1 (@ 1 \text{ bar}) = -289.47 \text{ kJ/(kmol} \cdot \text{K)}$$

Solving for T_2^{ideal} so that $S_2^{\text{ideal}} (@ 11.436 \text{ bar}) = S_1 (@ 1 \text{ bar})$

$$\text{we obtain } T_2^{\text{ideal}} = 49.26 \text{ } ^\circ\text{C} \quad H_2^{\text{ideal}} (@ 11.436 \text{ bar}) = -102,061.12 \frac{\text{kJ}}{\text{kmol}}$$

Therefore the ideal work is $W^{\text{ideal}} = (H_2 - H_1) \times \text{flow rate} \Rightarrow$

$$W^{\text{ideal}} = \underline{1.84 \text{ MW}} \quad \text{To calculate the real requirements}$$

we assume an actual temperature $T_2 > 49.26 \text{ } ^\circ\text{C}$ and

recalculate the real work W^{real} . We now apply

goalseek on the temperature T_2 so that $\frac{W^{\text{real}}}{W^{\text{ideal}}} = 0.75$

$$\text{That gives us } T_2 = 62 \text{ } ^\circ\text{C} \text{ and } W^{\text{real}} = \underline{2454.4 \text{ kW}}$$

We can now calculate the condenser duty

$$\dot{Q}_C = \dot{Q}_{\text{CHILLER}} + \dot{W}_{\text{REAL}} \Rightarrow \dot{Q}_C = 2900.55 + 2454.4 \Rightarrow$$

$$\dot{Q}_C = \underline{\underline{5354.96 \text{ kW}}}$$

The coefficient of performance is defined as

$$\text{COP} = \frac{\dot{Q}_{\text{CHILLER}}}{\dot{W}_{\text{REAL}}} = \frac{5354.96}{2454.4} \Rightarrow \text{COP} = \underline{\underline{2.18}}$$

The maximum COP is for reverse Carnot cycle between $T_1 = 35^\circ\text{C}$ and $T_2 = -42.15^\circ\text{C}$

$$\text{COP} = \frac{1}{1 - \frac{T_2}{T_1}} = \frac{T_1}{T_1 - T_2} = \frac{35 + 273.15}{35 - (-42.15)} \Rightarrow \text{COP}_{\text{max}} = \underline{\underline{4.0}}$$

Case 2, If the compression is polytropic, then we need to recalculate the power requirements. The ideal work calculation does not change, however for the real work we must use the polytropic coefficient defined by

$$\eta = \frac{\left(r^{\frac{n-1}{n}} - 1\right) \frac{n(k-1)}{k(n-1)} \frac{W_{\text{Ideal}}}{W}}{\left(r^{\frac{k-1}{k}} - 1\right)} \quad r = \frac{P_{\text{out}}}{P_{\text{in}}} \quad t = \frac{Q_{\text{out}}}{Q_{\text{in}}} \quad t^{\text{ideal}} = \frac{Q_{\text{out}}^{\text{ideal}}}{Q_{\text{in}}}$$

$$\eta = \frac{\ln r}{\ln t} \quad k = \frac{\ln r}{\ln t^{\text{ideal}}}$$

The final results are $T_2 = 66^\circ\text{C}$ and $\dot{W} = \underline{\underline{2564.8 \text{ kW}}}$